DroneVPS — Project Status

Marker-free GNSS-denied navigation for a 4.0 kg hexacopter. Target: 100-150 m AGL, 3-5 km out-and-back, no GNSS, no ground marker. Twenty-one flights across 20-23 October. The 60 m stall is gone, derotation moved to the camera gyro, and both mono cameras now run optical flow. Position held 201 s on the VPS at 67 m average, and the best flight's error against the raw receiver is 3.88 m. What still ends a window is velocity.

The 60 m stall was the matcher's model loading, and it is gone.

The XFeat model was built lazily, so the first call landed on the first frame above 60 m - 2.35 s warm, 4.95 s cold, with the GIL held. It cost the flow loop 2.6 s in EVERY flight at exactly that height: 232 at t 218.0, 233 at t 257.2, 236 at t 321.0. It is built at worker start now, on the ground. Flight 237 crossed 60 m with no gap at all.

Derotation was never running, and now it is.

derot_src read 'fc' on all 15019 frames of flight 236 while the FC's own per-frame delta measured 0.000 - the FC arrives at 4 Hz against a 30 Hz loop. The camera gyro saw 0.065 deg of roll and 0.120 of pitch per frame and was being discarded. It carries 100% of frames now, with the sign from the constants rather than re-measured in flight.

Two cameras now run flow, and they agree to 0.05 px.

The mount came down 2-3 cm, the landing leg left CAM_B's frame, and CAM_B's frames reach the CPU for the first time - they went into StereoDepth on the VPU and never crossed XLink. Each camera has its own intrinsics. Tracks p50 211 -> 398. Their disagreement is the first bound this project has had on the flow's own noise instead of an inference.

Overall
86%

An item is 100% only when it has been verified in flight. 84% on 22 Oct. Up two points: the 60 m stall, derotation and t them has flown; dual flow and the 1 Hz map period have not.

The switch could swallow the pilot's command, and did.

gps_rc_switch skipped the write when its cached belief already matched. One wrong read-back - 1 in 64 this session, at 10:31:51 - became a wrong aircraft: flight 233 landed on the VPS with the transmitter asking for GPS, and flight 237 ignored a VPS request for 119 s. The cache no longer gates the write, every write is read back and retried, and the refusal that kept the aircraft on GPS when the node was dead is gone - switch means switch.

1. What changed since the 22 October report

Everything below is in the tree now. 'Flown' means it was in the air on flight 237 or later; nothing marked unflown has been tested outside the bench.

Matcher model warmed on the ground	**new**	FLOWN (237). 2.4-2.6 s at worker start. The 2.6 s gap at 60 m is gone - 237 crossed it with no loop gap, against one in every earlier flight.
Matcher runs below 60 m, result discarded	**new**	FLOWN (237). Keeps the path warm and proves in the air that the matcher is alive before it is needed. Nothing below min_alt_m reaches the fuser, which is unchanged.
Gyro derotation, 100% of frames	**new**	FLOWN (237). Sign from the constants (-14.80 / -13.82), not from TiltValidator, which picks it by matching rotation when what matters is translation. The validator still reports.
Switch: no cache skip, verify + retry	**new**	UNFLOWN. Queue drained before every read, three retries, SOURCE FAILED on the radio. 63 of 64 commands this session were already correct; the one that was not cost a landing.
Switch means switch	**new**	UNFLOWN. VPS REFUSED removed. Ask for the VPS with the node dead and you get the VPS and an ALT_HOLD, not a receiver nobody asked for.
Handover carries nothing	**new**	UNFLOWN. The velocity bridge had no reset at all, so its integrated velocity and learned bias crossed every switch. Position was always reset; this is the other half.
Dual flow, both mono cameras	**new**	UNFLOWN. CAM_B has its own tracker and its own K/D. Weighted by track count. Tracks p50 211 -> 398, cameras agree to 0.05 px, 56% of frames pair at 25 fps.
mono_fps 30 -> 25	**new**	UNFLOWN. 30 asked for delivered 28.5 with 22% of frames late once the second stream was added. 25 delivers 23.8 with 12.8% late, and pairing rose 42% -> 56%.
fix_period 2.0 -> 1.0	**new**	UNFLOWN, and the measurement disagrees - see section 4. At the pilot's instruction, to be flown. It doubles what the map can pull back: 0.24 m/s against 0.12.
Camera mount 2-3 cm lower	**new**	FLOWN. The leg left CAM_B's frame - it sat at x 112-179, y 728-800, 0.47% of the frame. No recalibration needed: gravity still lands on cam_z at 99.8%, so the mount did not rotate.

2. Position estimation

Optical flow	**95%**	Unchanged in the air. Two cameras now, averaged by track count; the second has not flown.
Gyro derotation	**90%**	Was 85% and the item was wrong: it was not running at all. 100% of frames now carry the camera gyro. Flown on 237.
Camera IMU configuration	**95%**	169 Hz delivered on gyro, accel and rotation-vector with both camera streams up. 400 requested.
Mount alignment	**92%**	Mount lowered 2-3 cm. Gravity on cam_z 99.8%, IMU-to-CAM_C 179.6 deg - both unchanged, so the move was a translation.
Map matching	**85%**	39 fixes on 237 against 1 on 236, and the fixes kept coming to within 5.5 s of the failsafe. Error against the receiver 1.72 m at 60-75 m.
Sensor fusion	**75%**	Unchanged. max_correction_m 0.5.
Velocity to the FC	**60%**	Unchanged, and it is still what ends every window. Both of 237's failsafes are the sent speed reaching the 6 m/s cap.
Heading / yaw	**55%**	Unchanged.
Position injection as GPS2	**95%**	Measured clean on 237: on the VPS the EKF sits 1.22 m from ours and 4.30 m from the M9N; on GPS1, 0.13 m from the M9N.

3. Control and the source switch

RC source switch	**92%**	Was 85%. Cache skip, read-back race and the health refusal all fixed. Unflown.
Holding position on the VPS	**72%**	Was 70%. 201 s at 67 m average on 237, one failsafe, position p50 3.88 m against the raw receiver.
Autonomous mission	**40%**	Not attempted on the VPS since the rebuild.
Mission speed ceiling	**70%**	Unchanged.

4. Measured on 23 October

The 60 m stall, located	**new**	2.35 s warm / 4.95 s cold model load. One loop gap per flight, 2.62-2.67 s, within 0.1 s of first crossing 60 m, in all three flights.
fix_period, scored honestly	**new**	Against GPS_RAW_INT: 2.0 s gives 1.35 m over all VPS rows and 2.76 m above 100 m; 1.0 s gives 5.33 and 11.16. The sweep that favoured 1.0 was scored against the map's own fixes.
The map does NOT improve with height	**new**	149 fixes, two flights: 1.72 m at 60-75 m, 4.61 m at 105-130 m. Altitude against error correlates +0.31. Inliers DO rise (22 -> 35) - more features, each worth more metres.
Why the map could not save 237	**new**	0.5 m per fix every 4.1 s is 0.12 m/s of correction. The error grew 3.19 -> 11.48 m over the same minute, 0.14 m/s. It was losing a race, not missing.
Both failsafes on 237	**new**	t 246.4 at 88 m: tracks 231 -> 80, flow stale, sent speed to the 6 m/s cap, XKF4.SV 0.66 -> 2.87. t 401.9 at 65 m: tracks fine at 187, sent speed still ran 0 -> 6.00.
Stick input is what broke the hold	**new**	Pitch stick 1495 -> 1045 from t 212 to 238. SV was 0.5-1.0 while centred and touched 1.6 under stick. The ALT_HOLD came 4 s AFTER the stick returned to centre.
Flow scale is the accuracy problem	**new**	Slope against the receiver swings 0.51 at low rotation to 2.68 at high, correlation 0.27-0.46. Shared between both cameras, so averaging cannot touch it.
Jetson load, everything running	**new**	CPU 72% p50 / 77% p90, drone-vps 2.7 of 6 cores, 567 MB, 70 C steady. Router and switch 1.8% each. No service is overloaded; the margin is 23%.

5. Next

Fly it. Nine changes have never been in the air.	**new**	Dual flow, 25 fps, fix_period 1.0, the switch fixes, switch-means-switch and the handover reset. The switch ones are safety; the flow ones change what the FC is told.
Velocity is still the wall.	**new**	Both of 237's failsafes are the sent speed reaching 6 m/s. Above 60 m the velocity error is p90 3.06 m/s against 0.76 below, and every tau from 0.35 to 0.08 lands within 14% of that. Smoothing is not the lever.
The flow's scale, measured against the receiver.	**new**	Slope 0.51 to 2.68 with rotation. Until this is one number the map is correcting an estimate whose gain is unknown, and no map period can fix that.
Declared accuracy above 60 m.	**new**	On 237 it is nearly honest - 3.40 declared against 3.88 real. On 236 it was 4.80 against 29.22. The EKF tests its innovation against this number, so it has to stay honest as the rest improves.
The Lua switch on the flight controller.	**new**	SCR_ENABLE is 1 and RC6_OPTION is 0. A script on the FC would read the channel and set GPS_PRIMARY itself, so the switch works with the Jetson off. The pilot's rule: no automatic source change, ever - if the VPS dies the answer is ALT_HOLD, not GNSS.
The clock, at the system level.	**new**	Flight 234 was lost to it: a 1970 clock makes the GPS week negative, it will not pack into the uint16 field, and the node died 17 times. 64 of 68 CSV columns and every test pass, but a position source must not exit over a timestamp.

What gates the mission, lowest first: 3-5 km out-and-back 15%, autonomous mission 40%, heading/yaw 55%, velocity to the flight controller 60%, altitude ceiling 60%, mission speed ceiling 70%, holding position on the VPS 72%, sensor fusion 75%, map matching 85%. The aircraft held 201 s on the VPS at 67 m on 23 October and its position was good to 3.88 m against the raw receiver, which is the best this project has measured. The 60 m stall and the derotation hole are closed. What is left is the velocity the flight controller is given, the flow's own scale, and distance.